

SOC Analyst in Social Science

Student Name: Carlos Morales

School of Cybersecurity, Old Dominion University

CYSE 201S: Cybersecurity and Social Sciences

Instructor Name: Diwakar Yalpi

Date: 04/18/2026

Cybersecurity is not solely a technical discipline focused on protecting systems and networks; it is also deeply rooted in human behavior and social processes. Security Operations Center (SOC) analysts rely on an understanding of how people think, react, and make decisions in order to identify threats, respond effectively to incidents, and maintain system integrity. This paper argues that the effectiveness of cybersecurity depends on interpreting human behavior, applying SOC operational principles, and recognizing broader societal issues such as marginalization that shape digital risk.

Cybersecurity is a constantly evolving field dedicated to defending digital infrastructure and data from threats such as unauthorized access, malware, and cyber intrusion. Within this field, SOC analysts play a central role by monitoring security environments, investigating alerts, and responding to incidents as they occur. While technical expertise is essential, success in this profession also depends on social science foundations, including behavioral psychology and decision-making processes. This essay examines how social science concepts influence cybersecurity practice, how SOC principles are applied in real-world environments, and how cybersecurity intersects with inequality and marginalization in society.

Human behavior is a major factor in cybersecurity because many attacks target people rather than systems. Techniques such as phishing, impersonation, and social engineering are designed to manipulate emotions like trust, urgency, and fear in order to influence decisions. These methods are effective because they exploit consistent patterns in human judgment, especially when individuals are under pressure or uncertainty, rather than exploiting purely technical vulnerabilities. As a result, cybersecurity requires insight into behavioral responses and cognitive processes in high-stress situations, which aligns closely with psychological research on decision-making under uncertainty. Within SOC environments, analysts process large volumes

of alerts and must determine which events represent genuine threats. Reeves and Ashenden (2023) highlight that “the role of a SOC analyst is cognitively demanding,” emphasizing how continuous evaluation of threats places heavy mental strain on analysts. When cognitive load becomes excessive, it can reduce attention and decision accuracy, increasing the likelihood of missed or misinterpreted threats. From this perspective, cybersecurity outcomes are strongly shaped by human cognitive limitations such as stress, attention span, and memory capacity.

Cybersecurity communication systems can also be deliberately manipulated by attackers. Phishing attacks, for example, disrupt normal communication processes by imitating trusted sources and altering message content to mislead recipients, effectively inserting intentional “noise” into information channels (Dietl et al., 2023). Within SOC operations, even highly technical security alerts may lead to incorrect conclusions if they are not clearly communicated or properly contextualized during escalation. These communication failures can slow response times or lead to incorrect threat prioritization, increasing organizational vulnerability. This demonstrates that cybersecurity performance relies not only on technical infrastructure but also on how effectively information is exchanged and interpreted under pressure. In addition, team dynamics and psychological safety play a significant role in SOC effectiveness. Psychological safety describes an environment where individuals feel secure enough to express concerns, ask questions, and report issues without fear of negative consequences. In SOC teams, this is particularly important because analysts must collaborate and share interpretations quickly when responding to uncertain threats. When psychological safety is strong, team members are more likely to raise concerns early, improving collective awareness and reducing the likelihood of errors. Overall, SOC effectiveness depends on coordinated communication, collaboration, and decision-making under time-sensitive and cognitively demanding conditions.

Marginalization in cybersecurity reflects broader structural inequalities in access to technology, education, and digital protection. From a social science perspective, exposure to cyber threats is unevenly distributed, with individuals and communities that have fewer economic or educational resources being more vulnerable to risks such as phishing scams, identity theft, and financial fraud. These disparities show that cybersecurity is shaped not only by technical defenses but also by social and economic structures that influence access to information and protection. Research on technological development highlights this imbalance, noting that “public institutions, policies and financial resources pervasively privilege” certain areas of innovation, which creates gaps between societal needs and resource allocation (Wilson et al., 2012). As a result, populations with the greatest exposure to cyber risks often have the least access to protective tools and cybersecurity education. These inequalities are also reflected within SOC environments. Analysts must prioritize alerts based on severity, relevance, and available resources, which means that some systems or users may receive less immediate attention if their risks are not clearly flagged in monitoring frameworks. From a social science standpoint, this demonstrates how patterns of inequality and resource distribution influence cybersecurity decision-making at both the organizational and societal levels.

In conclusion, the integration of social science concepts into cybersecurity demonstrates that security is not only a technical challenge but also a behavioral and structural one. Improving cybersecurity therefore requires more than advanced technologies; it also depends on understanding how people make decisions under pressure, strengthening communication and teamwork within security operations, and addressing inequalities that shape access to digital protection.

Works Cited

Dietl JE, Derksen C, Keller FM, Lippke S. Interdisciplinary and interprofessional communication intervention: How psychological safety fosters communication and increases patient safety. *Front Psychol.* 2023 Jun 15; 14:1164288. doi: 10.3389/fpsyg.2023.1164288. PMID: 37397302; PMCID: PMC10310961.

Reeves, Andrew, and Debi Ashenden. "Understanding Decision Making in Security Operations Centres: Building the Case for Cyber Deception Technology." *Frontiers in Psychology* [Switzerland], vol. 14, May 2023, p. 1165705, <https://doi.org/10.3389/fpsyg.2023.1165705>.

Wilson, Charlie, et al. "Marginalization of End-Use Technologies in Energy Innovation for Climate Protection." *Nature Climate Change* [London], vol. 2, no. 11, November 2012, pp. 780–88, <https://doi.org/10.1038/nclimate1576>.